Rank-27 Current Collection Confirmation

Tactix Development Project

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Outcome

The accepted Vivify and Congruence deferrals have reduced the exact Rank-27 common-prefix collection population from the historical 36 calls to 22. A fresh current-source observer reproduces every accepted semantic statistic and confirms that the remaining current-Kissat gap is collection frequency, not per-call compact/forward throughput.

This is a localization closure, not a new solver repair. It supplies no contradictory evidence to reopen the rejected Substitute call-level deferral, stable-span Vivify representation, fused compact-forward traversal, or quarantined arena-prepass prototype. Broad-search convergence therefore moves outside those closed arms.

1 Exact current observation

Single-solver job `1650175` ran on node018 with 12 logical CPUs reserved, exactly 48,000 MiB scheduler memory, and a 16 GiB process cap. It stopped intentionally UNKNOWN at conflict 9450. After removing input hashes, timing, observer, and perf-wrapper rows, its complete semantic output has SHA-256 `3b87bb36da711bcef3ca877f3922e9fa5f8a3c6ae9d638c1ae18d4965bd92c5f`, byte-identical to the accepted common-state authority.

The immutable observer pin has SHA-256 `12df42b9b4e9d851145986b4fd3711ce21db7ac94419e90ee5e6839ad495fd25`. Its instrumented source is frozen separately; the accepted worktree source was restored byte-for-byte before submission.

2 Population comparison

Metric	Current Tactix	Current Kissat	Ratio
Collection calls	22	7	3.143
Input watch entries	59,408,930	19,062,574	3.1165
Entries per call	2,700,406	2,723,225	0.9916
Surviving large entries	5,759,788	5,759,788	1.0000

Tactix additionally scans 25,877,700 arena records / 211,924,478 words, rewatches 9,914,796 clauses / 31,554,270 literals, and rebuilds 31,322,787 dense occurrences. The current-Kissat values remain the accepted job `1647659` conservative conflict-10000 envelope; Kissat was not rerun.

The equal per-call entry population and equal surviving-large population make the owner precise: Tactix invokes equivalent maintenance about three times as often. The earlier bounded fused traversal remains correctly rejected.

3 Profile and disposition

perf captured 2,643 samples with zero loss. Shares are 2.01% in `kernel_collect_garbage`, 1.76% in stable watch compaction, 1.66% in dense occurrence rebuilding, and only 0.33% in the second forwarding walk. The staged CNF was deleted; no proof, model, checker, certificate, or scratch artifact was created.

The accepted source already removed the portable Congruence-owned calls. The remaining known call-level alternatives are closed by exact negative, neutral, or crash evidence. The next action is a current-source broad-search call-graph profile against the retained current-Kissat profile, excluding those closed owners, followed by source-backed localization of the first remaining equal- work cost or weaker-state gap.